Complex Analysis

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Chapter 1

Introduction

Ask someone what the square root of 4 is, and they will not hesitate: 2. Ask for the square root of -1, and something interesting happens. Most people pause, then say some version of the same thing: *that doesn't exist.*

They are not wrong to hesitate. For over a thousand years, some of the sharpest minds in mathematics agreed with them. A number that, when multiplied by itself, gives a negative result seemed less like a number and more like a contradiction a hole in the number line where nothing should be allowed to stand. When the number finally forced its way into mathematics, it did not arrive through some philosopher's curiosity about the impossible. It arrived because mathematicians doing ordinary, practical work solving cubic equations with three perfectly good, perfectly real solutions found that the only way to reach those real answers was to pass, briefly, through square roots of negative numbers. The imaginary was not a detour around the real. It turned out to be sewn into it.

This book is about that number, and the strange, beautiful mathematics that grew up around it.

I want to be honest about something before we start: I did not always find complex numbers beautiful. For a long time they felt like a trick a symbol, i, invented so that certain equations would stop being unsolvable, defined by little more than the rule $i^2 = -1$ and otherwise left to behave like an ordinary number. It took years of study, first as a student and now as a researcher in complex analysis, before I understood what was actually being offered: not a patch, but a second dimension. The moment you accept i, the number line which had felt complete, self-sufficient, the whole of arithmetic turns out to have been only one axis of a much larger plane. Numbers stop being points on a line and become points in a plane, and multiplication, of all things, becomes rotation.

If you grew up here in Sri Lanka, you may already have an intuition for this kind of shift in perspective, even if no one called it mathematics. Think of the geometric repetition in a batik pattern, or the way a *rasam pot's* shadow rotates through the day, or the careful symmetry worked into a *vesak koodu.* Rotation, reflection, repetition with variation these are not decorations on top of mathematics. They are mathematics, and complex numbers turn out to be the precise language for describing them. By the time we reach the middle of this book, you will see multiplication by i the way you already see a quarter-turn of a lantern: not as an abstract operation, but as a motion you can picture.

We will take the historical route, because I think it is the honest one. Real numbers themselves the ones you already trust were not always on solid ground. The Pythagoreans were shaken to discover that $\sqrt{2}$ could not be written as a fraction. Indian mathematicians, including Brahmagupta in the seventh century, were the first to write down clear arithmetic rules for negative numbers and zero, centuries before Europe caught up. The Kerala school, working in South India in the 14th to 16th centuries, developed infinite series for trigonometric functions long before calculus had a name in the West. None of this was inevitable or obvious at the time. Every number system we now take for granted negative, irrational, real, and eventually imaginary was once controversial, resisted, or simply unnamed.

So we will not start with a definition of i and a list of rules to memorize. We will start with the crisis that made it necessary, follow the mathematicians who fought over what to make of it, and only then turn to the elegant plane, the rotations, and the surprising places electrical engineering, quantum mechanics, the shape of fractals where this once-imaginary number turns out to be indispensable.

By the end, I hope $\sqrt{-1}$ will feel less like an exception to arithmetic and more like the moment arithmetic finally became honest about its own shape.

1.0.1 Why Complex Numbers

Chapter 2

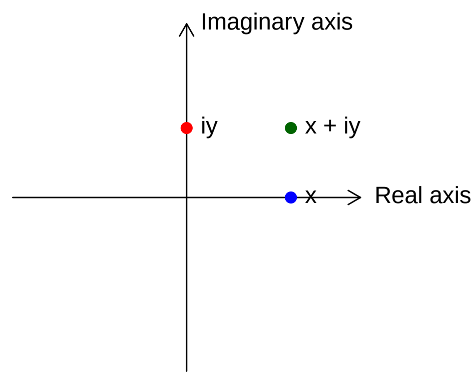
Complex Numbers

A complex number is defined as $z = x + iy$, where $x, y \in \mathbb{R}$. Here, x represents the **real part** ($\operatorname{Re} z$) and y represents the **imaginary part** ($\operatorname{Im} z$).

The set of all such numbers, $\mathbb{C}$, constitutes the complex plane. Since $\mathbb{R} \subset \mathbb{C}$, real numbers are identified as complex numbers where $\operatorname{Im} z = 0$.

There exists a bijective correspondence between $\mathbb{C}$ and the Euclidean plane $\mathbb{R}^2$, expressed as:

$$z = x + iy \longleftrightarrow (x, y)$$



In this geometric representation:

- The **real axis** corresponds to the x-axis.
- The **imaginary axis** ($i\mathbb{R}$), containing purely imaginary numbers of the form iy, corresponds to the y-axis.

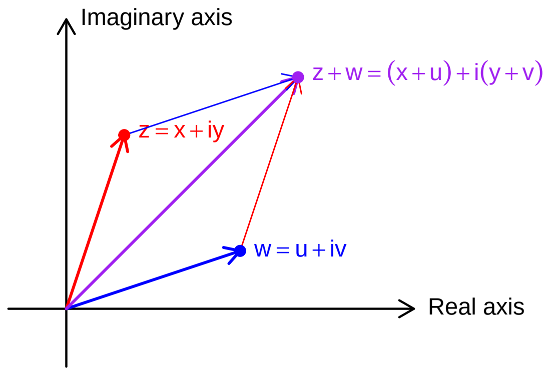
2.0.1 Basic Definitions and Algebra

- A complex number has the form $z = a + ib$ where a, b are real numbers
- a is the **real part**: $\mathrm{Re}(z) = a$
- b is the **imaginary part**: $\mathrm{Im}(z) = b$
- A number is **purely imaginary** if $a = 0$
- A number is **real** if $b = 0$
- Zero is the only number that is both real and purely imaginary

2.0.2 Arithmetic Operations

2.0.2.1 Addition

$z + w = (x + iy) + (u + iv) = (x + u) + i(y + v)$



2.0.2.2 Subtraction

$z - w = (x + iy) - (u + iv) = (x - u) + i(y - v)$

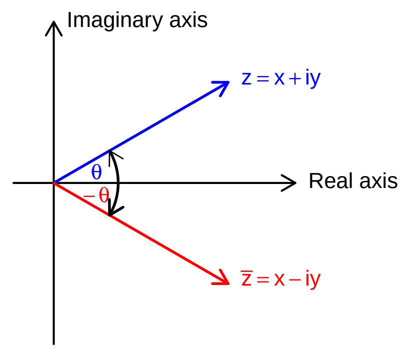
2.0.2.3 Multiplication

$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$

2.0.2.4 Division

$$\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = \frac{ac+bd}{c^2+d^2} + i\frac{bc-ad}{c^2+d^2}$$

2.0.3 Complex Conjugate



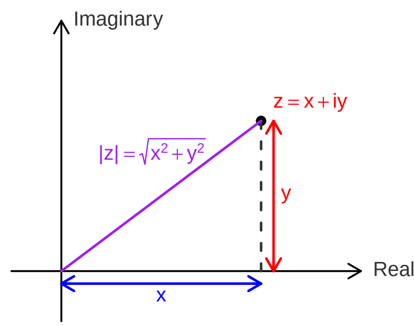
- The conjugate of $z = a + ib$ is $\bar{z} = a - ib$

- Properties: For all $z, w \in \mathbb{C}$:

  - $\overline{z+w} = \bar{z} + \bar{w}$
  - $\overline{zw} = \bar{z}\,\bar{w}$
  - $\overline{(z/w)} = \frac{\bar{z}}{\bar{w}},\ w \neq 0$
  - $\frac{1}{z} = \frac{\bar{z}}{|z|^2},\ z \neq 0$
  - z is real $\Longleftrightarrow z = \bar{z}$
  - z is purely imaginary $\Longleftrightarrow z = -\bar{z}$

2.0.4 Absolute Value (Modulus)

- The absolute value or modulus of $z = a + ib$ is:



- $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$

- Properties:
  - $|z| \geq 0$ for all z, with equality if and only if $z = 0$
  - $|zw|^2 = (zw)(\overline{zw}) = (z\bar{z})(w\bar{w}) = |z|^2|w|^2$

  - $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
  - $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ for $z_2 \neq 0$
  - $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle inequality)
  - $||z_1| - |z_2|| \leq |z_1 - z_2|$

2.0.5 Important Inequalities

1. **Triangle Inequality**: $|z_1 + z_2| \leq |z_1| + |z_2|$

   - Equality holds if and only if $\frac{z_1}{z_2} > 0$ (i.e., the ratio is positive real)

2. $-|z| \leq \mathrm{Re}(z) \leq |z|$

   - Equality on right when z is positive real

3. $-|z| \leq \mathrm{Im}(z) \leq |z|$

4. **Cauchy's Inequality**: $|a_1b_1 + a_2b_2 + ... + a_nb_n|^2 \leq (|a_1|^2 + ... + |a_n|^2)(|b_1|^2 + ... + |b_n|^2)$

   - Equality holds if and only if the a_i are proportional to the b_i

2.0.6 Fundamental Theorem of Algebra

A complex polynomial of degree $n \geq 0$ is a function of the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \qquad z \in \mathbb{C},$$

where $a_0, a_1, \ldots, a_n$ are complex numbers and $a_n \neq 0$.

A fundamental property of the complex numbers is that every polynomial with complex coefficients can be factored completely into linear factors.

Theorem 2.1 (Fundamental Theorem of Algebra)**.** *Every complex polynomial $p(z)$ of degree $n \geq 1$ admits a factorization*

$$p(z) = c\,(z-\zeta_1)^{m_1}(z-\zeta_2)^{m_2}\cdots(z-\zeta_k)^{m_k},$$

where the numbers $\zeta_1, \ldots, \zeta_k$ are distinct complex roots of p, each multiplicity satisfies $m_j \geq 1$, and $m_1 + m_2 + \cdots + m_k = n$. This factorization is ***unique****, up to a permutation of the factors.*

Proof. Omitted. Multiple complete and rigorous proofs are available at ProofWiki. ☐

The uniqueness of the factorization is straightforward. The points $\zeta_1, \ldots, \zeta_k$ are uniquely determined as the zeros of $p(z)$, that is, the solutions of $p(z) = 0$. Each multiplicity m_j is the unique integer $m \geq 1$ for which $p(z)$ can be written in the form $(z-\zeta_j)^m q(z)$ with $q(\zeta_j) \neq 0$.

To prove that such a factorization exists, one uses induction on the degree n of the polynomial. The key step is to show that $p(z)$ has at least one root ζ_1. Once a root is found, the polynomial can be factored as $p(z) = (z-\zeta_1)q(z)$, where $q(z)$ has degree $n-1$. By the induction hypothesis, $q(z)$ can be factored into linear factors, which completes the factorization of $p(z)$. Thus, the Fundamental Theorem of Algebra is equivalent to the statement that every complex polynomial of degree $n \geq 1$ has at least one zero.

2.1 Geometric Representation

2.1.1 The Complex Plane

- A complex number $z = a + ib$ corresponds to the point (a, b) in the plane
- The x-axis is called the **real axis**
- The y-axis is called the **imaginary axis**
- The plane is called the **complex plane**

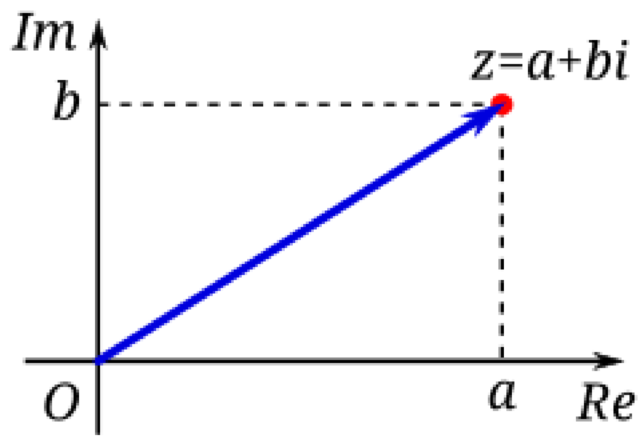


Figure 2.1: Image Source: Wikimedia

2.1.2 Vector Interpretation

- Complex numbers can be represented as vectors from the origin
- Vector addition corresponds to complex addition
- The length of the vector is $|z|$
- The distance between points z_1 and z_2 is $|z_1 - z_2|$

2.1.3 Polar Form

- A complex number can be written in **polar form**:
  - $z = r(\cos\theta + i\sin\theta)$ where $r = |z|$ and $\theta = \arg(z)$
  - r is the modulus: $r = |z| = \sqrt{a^2 + b^2}$
  - θ is the **argument** or **amplitude**: $\theta = \arg(z)$
- Properties of polar form:
  - For $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$:
  - $z_1 z_2 = r_1 r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$
  - $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
  - $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

2.1.4 De Moivre's Formula

- For $z = r(\cos\theta + i\sin\theta)$:
  - $z^n = r^n(\cos n\theta + i\sin n\theta)$
- For $r = 1$:

- $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

2.1.5 Roots of Complex Numbers

- The n-th roots of $z = r(\cos\theta + i\sin\theta)$ are:
  - $\sqrt[n]{z} = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2\pi k}{n}\right) + i\sin\left(\frac{\theta+2\pi k}{n}\right)\right]$ for $k = 0, 1, 2, \dots, n-1$
- There are n distinct n-th roots of any non-zero complex number
- These roots have equal moduli and are equally spaced around a circle
- For $z = 1$, the roots are called the **n-th roots of unity**

2.1.6 Equations in the Complex Plane

- Circle with center a and radius r: $|z - a| = r$
  - In algebraic form: $(z-a)(\bar{z}-\bar{a}) = r^2$
- Straight line: $z = a + bt$ where t is a real parameter
  - Two lines are parallel if b' is a real multiple of b
  - Lines are orthogonal if $\frac{b'}{b}$ is purely imaginary
  - The angle between lines is $\arg\frac{b'}{b}$

2.1.7 Stereographic Projection and the Riemann Sphere

The extended complex plane is the complex plane together with the point at infinity. We denote the it by $C^* = c \cup \{\infty\}$. - The extended complex plane includes ∞ and can be represented on a unit sphere - Under stereographic projection, the point (x_1, x_2, x_3) on the sphere corresponds to: - $z = \frac{x_1+ix_2}{1-x_3}$

- The north pole $(0, 0, 1)$ corresponds to ∞
- Properties:
  - Circles on the sphere correspond to circles or lines in the plane
  - The distance formula: $d(z, z') = \frac{2|z-z'|}{\sqrt{(1+|z|^2)(1+|z'|^2)}}$
  - The sphere is often called the **Riemann sphere**

2.1.8 Key Formulas and Results

1. $z\bar{z} = |z|^2 = a^2 + b^2$
2. $\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{a-ib}{a^2+b^2}$
3. $z^n = r^n(\cos n\theta + i\sin n\theta)$

4. $\sqrt[n]{z} = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2\pi k}{n}\right) + i\sin\left(\frac{\theta+2\pi k}{n}\right)\right]$ where $k = 0, 1, \ldots, n-1$
5. Triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$
6. Cauchy's inequality: $\left|\sum_{i=1}^{n} a_i \bar{b}_i\right|^2 \leq \left(\sum_{i=1}^{n} |a_i|^2\right)\left(\sum_{i=1}^{n} |b_i|^2\right)$
7. The equation of a circle: $|z - a| = r$
8. The roots of unity: ω^k where $\omega = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ and $k = 0, 1, \ldots, n-1$

2.2 Exercise

1. Find the values of

$$(1+2i)^3, \quad \frac{5}{-3+4i}, \quad \left(\frac{2+i}{3-2i}\right)^2, \quad (1+i)^n + (1-i)^n.$$

2. If $z = x+iy$ (where x and y are real numbers), find the real and imaginary parts of:

$$z^4, \quad \frac{1}{z}, \quad \frac{z-1}{z+1}, \quad \frac{1}{z^2}.$$

3. Show that:

$$\left(\frac{-1 \pm i\sqrt{3}}{2}\right)^3 = 1 \quad \text{and} \quad \left(\frac{\pm 1 \pm i\sqrt{3}}{2}\right)^6 = 1$$

for all combinations of signs.

Chapter 3

Green functions

Let G be a reigon and $a \in G$. Green functions G with singularity at a, the unique function $g_a : G \setminus \{0\}$

1. g_a is harmonisc on \$G {a}.
2. $g_a(z) + \log(|z-a|)$ is harmonic[1] on an open neighbourhood of a.
3. $\lim_{z \to w} g_a(z) = 0 \quad \forall w \in \partial_\infty G$.

::: {.remark} $g_a(z) > 0$ for all $z \in G \setminus \{a\}$ because $g_a(z) + \log(|z-a|)$ has contious externsion to u and \$lim$_{z \to w} g_a(z) = -\infty$ :::

We have $g_a(z)0$ on $|z-a| = r$ for r small enough (then there exist R such that its positive $\forall r < R$) Given $r < R$ such that z_0 is $|z-a| = r$. Apply max principle on $G \setminus \{z : |z-a| \leq r\}$. Since $g_a(z) \geq 0$ or \$\$

[1]has an exteror to u. $(a \in u)$, u is open in G.

Chapter 4

Riemann Surfaces.

Definition 4.1 (Riemann surface)**.** A **Riemann surface** is a second-countable, Hausdorff topological space X equipped with a **complex atlas** $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ satisfying the following properties:

1. Each $U_\alpha \subseteq X$ is open.
2. Each map $\varphi_\alpha : U_\alpha \to V_\alpha$ is a homeomorphism onto an open set $V_\alpha \subseteq \mathbb{C}$.
3. The collection $\{U_\alpha\}$ covers X.
4. For every pair of charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the **transition map**

   $$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a **biholomorphism**.

Remark. **Terminology**: -"holomorphic functions" are functions which are complex differentiable on an open set in the complex plane. -A "biholomorphic map" is a holomorphic bijection whose inverse is holomorphic.

Definition 4.2. Let R_1 and R_2 be Riemann surfaces. A map $f : R_1 \to R_2$ is said to be **holomorphic** if for every pair of charts $(U, \varphi) \quad \text{on } R_1, \qquad (V, \psi) \quad \text{on } R_2$, the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic on its domain (which may be empty).

Suppose (U, φ), (U', φ') are charts on R_1, and (V, ψ), (V', ψ') are charts on R_2. On the overlap where all maps are defined, we have:

- the transition maps $\varphi' \circ \varphi^{-1}, \psi' \circ \psi^{-1}$ are **biholomorphisms** (by condition (5) in the definition of a complex atlas)

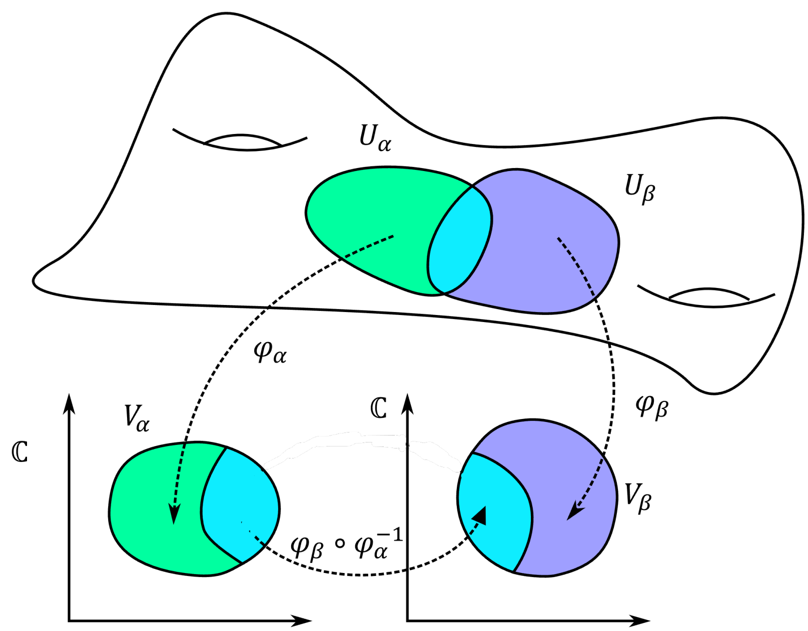


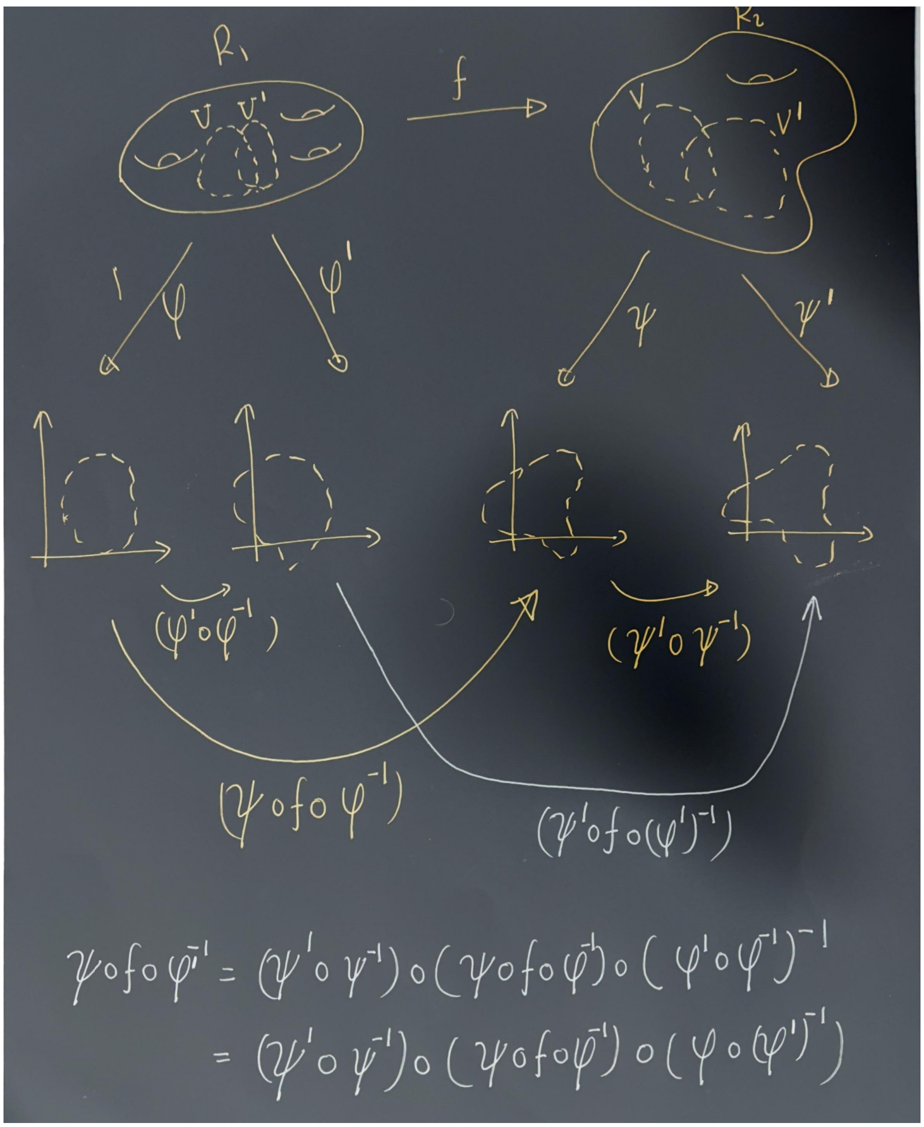
Figure 4.1: Two coordinate charts on a manifold (Image Credit: Wikipedia)

- therefore

$$\psi' \circ f \circ (\varphi')^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ (\varphi')^{-1})$$

  is holomorphic if and only if $\psi \circ f \circ \varphi^{-1}$ is holomorphic.

Thus the notion of holomorphicity does **not** depend on the choice of charts.



Exercise 4.1. Assume that for all $p \in R_1$ there is a chart (U, ϕ) such that $p \in U$ and a chart (V, ψ) such that $f(p) \in V$, so that $\psi \circ f \circ \phi^{-1}$ is holomorphic

(in the standard sense, as a complex analytic function on an open subset of the plane). Show that f is holomorphic.

Solution: We are given that for each $p \in R_1$, there exist some charts (U, ϕ) at p and (V, ψ) at $f(p)$ such that $\psi \circ f \circ \phi^{-1}$is holomorphic.

Now take any other charts (U_1, ϕ_1) at p and (V_1, ψ_1) at $f(p)$, with $f(U_1) \subset V_1$. On overlaps, the transition maps $\phi \circ \phi_1^{-1}$ and $\psi_1 \circ \psi^{-1}$ are biholomorphic (holomorphic with holomorphic inverses), because they are chart transition maps on a Riemann surface.

On the intersection where everything is defined, we can write $\psi_1 \circ f \circ \phi_1^{-1} = (\psi_1 \circ \psi^{-1}) \circ (\psi \circ f \circ \phi^{-1}) \circ (\phi \circ \phi_1^{-1})$.

- $\psi \circ f \circ \phi^{-1}$ is holomorphic by assumption.
- $\phi \circ \phi_1^{-1}$ and $\psi_1 \circ \psi^{-1}$ are holomorphic (indeed biholomorphic) transition maps.

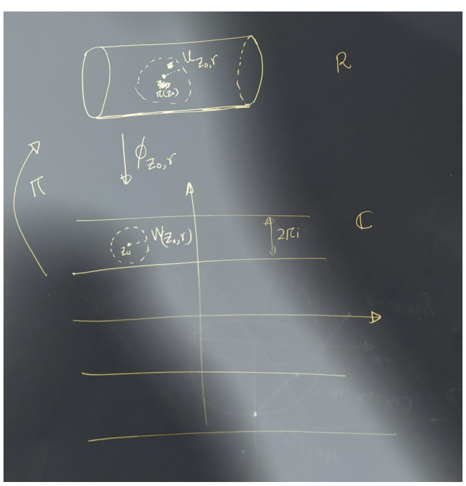
A composition of holomorphic maps is holomorphic, so $\psi_1 \circ f \circ \phi_1^{-1}$ is holomorphic.

Since this holds for any charts (U_1, ϕ_1), (V_1, ψ_1) around p and $f(p)$, it follows that f is holomorphic by the definition of a holomorphic map between Riemann surfaces.

Example 4.1. Define the equivalence relation on $\mathbb{C}$ by $z \sim w$ if and only if $z = w + 2\pi i n$ for some $n \in \mathbb{Z}$. Let $R = \mathbb{C}/\sim$ as a topological space, with projection map $\pi : \mathbb{C} \to R$.

We will define an atlas on R which will make it a Riemann surface. Define $W_{z_0,r} = \{ z \in \mathbb{C} : |z - z_0| < r \}$. Restrict to $r < \pi$ (for convenience). Let $U_{z_0,r} = \pi(W_{z_0,r})$.

Let $\varphi_{z_0,r} : U_{z_0,r} \to W_{z_0,r}$ be the unique right inverse of π on $U_{z_0,r}$ with image $W_{z_0,r}$.



Exercise 4.2. Show that for any z_1, z_2 and r_1, r_2 the map $\phi_{z_1,r_1} \circ \phi^{-1}_{z_2,r_2}$ is a translation, and in particular holomorphic. This shows that $\left\{(U_{z_0,r}, \phi_{z_0,r})\right\}_{\{z_0 \in \mathbb{C},\, r>0\}}$ is an atlas making R a Riemann surface.

4.1 Solution:

4.1.1 Functions defnied on the Riemann Surface.

4.1.1.1 Exp function

We know that $\exp z$ is a holomorphic function on $\mathbb{C}$ which is periodic in the sense that $\exp(z + 2\pi i n) = \exp(z)$ for all $n \in \mathbb{Z}$. Thus this passes to a well-defined function on R. Namely,

$$\mathrm{Exp} : R \longrightarrow \mathbb{C} \tag{4.1}$$

$$[z] \longmapsto \exp(z) \tag{4.2}$$

where $[z]$ is the equivalence class of z.

Exercise 4.3. Verify that this is well-defined.

Solution: To see that Exp is well-defined, assume $z \sim w$. Then there is an integer $n \in \mathbb{Z}$ such that $z = w + 2\pi i\, n$. Hence,

$$\mathrm{Exp}([z]) = e^z = e^{w+2\pi i n} = e^w\, e^{2\pi i n} = e^w = \mathrm{Exp}([w]).$$

Thus the value of $\mathrm{Exp}([z])$ depends only on the equivalence class $[z]$, so the function is well-defined.

Holomorphicity of Exp**.** To see that Exp is holomorphic, observe that for any chart $(U_{z_0,r}, \phi_{z_0,r})$ we have

$$\mathrm{Id} \circ \mathrm{Exp} \circ \phi^{-1}_{z_0,r}(z) = e^z,$$

which is holomorphic on $\mathbb{C}$.
Thus Exp is holomorphic by definition.

4.1.1.2 Sin function

The sine function. We may similarly define the function $\mathrm{Sin}([z]) = \sin z$, which is again well-defined and holomorphic.

Example 4.2. The complex plane $\mathbb{C}$ is a Riemann surface if we choose the atlas $\mathcal{A} = \{(U_{z_0,r}, \phi_{z_0,r})\}$. This was implicit in the previous example, where we composed by the identity map on the left.

Idea 1: Periodic functions are holomorphic functions on the Riemann surface obtained by taking quotients with respect to their periods.

Example 4.3. Let

$$R = \{(r, \theta) : r > 0,\ \theta \in \mathbb{R}\}.$$

This is trivially a topological space. We make it into a Riemann surface by defining charts as follows. For each $\theta_0 \in \mathbb{R}$, set

$$U_{\theta_0} = \{(r, \theta) : \theta_0 < \theta < \theta_0 + 2\pi,\ r > 0\},$$

and define

$$\phi_{\theta_0} : U_{\theta_0} \to \mathbb{C}, \qquad \phi_{\theta_0}(r, \theta) = re^{i\theta}.$$

Exercise 4.4. Show that for any pair θ_1, θ_2, the transition map

$$\phi_{\theta_1} \circ \phi_{\theta_2}^{-1}$$

is a biholomorphism.

4.1.1.3 LOG function

We then have a well-defined function

$$\mathrm{LOG} : R \to \mathbb{C}, \tag{4.3}$$
$$\mathrm{LOG}(r, \theta) = \log r + i\theta. \tag{4.4}$$

For any θ_0, the function

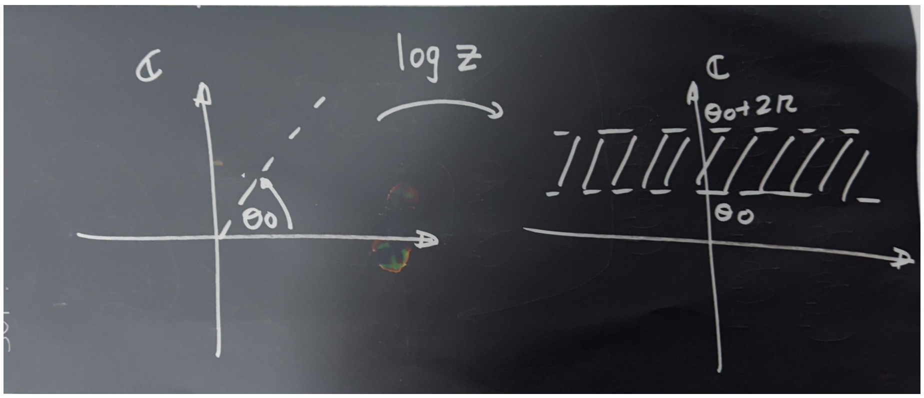
$$\mathrm{LOG} \circ \phi_{\theta_0}^{-1}$$

is a branch of the logarithm on the plane minus the slit $\arg z = \theta_0$. Explicitly,

$$\mathrm{LOG} \circ \phi_{\theta_0}^{-1}(z) = \log z,$$

where the right-hand side denotes the branch of the logarithm whose domain is the plane slit along the ray of argument θ_0, and whose values lie in the strip

$$\{w \in \mathbb{C} : \theta_0 < \Im(w) < \theta_0 + 2\pi\}.$$



4.2 The Riemann sphere

As a set, the **Riemann sphere** is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. To make this into a Riemann surface, we must specify a **topology** and an **atlas**.

A basis for the topology consists of:

1. Sets of the form $U \subset \mathbb{C}$, where U is open in $\mathbb{C}$ and $\infty \notin U$.
2. Sets of the form $\{\infty\} \cup (\mathbb{C} \setminus K)$, where $K \subset \mathbb{C}$ is compact.

Equivalently, a set $V \subset \widehat{\mathbb{C}}$ is open iff:

- If $p \in \mathbb{C}$, then there exists an open set $U \subset \mathbb{C}$ with $p \in U \subset V$.
- If $p = \infty$, then there exists a compact $K \subset \mathbb{C}$ such that $\{\infty\} \cup (\mathbb{C} \setminus K) \subset V$.

Exercise 4.5. Show that $\widehat{\mathbb{C}}$ is compact.

We define an atlas consisting of two charts:

$$U_1 = \widehat{\mathbb{C}} \setminus \{\infty\}, \qquad \varphi_1(z) = z.$$

$$U_2 = \widehat{\mathbb{C}} \setminus \{0\}, \qquad \varphi_2(z) = \frac{1}{z}.$$

The first chart and its inverse are clearly continuous.

To see that φ_2 is continuous, note that it maps $U_1 \cap U_2 = \widehat{\mathbb{C}} \setminus \{0, \infty\}$ bijectively onto $\mathbb{C} \setminus \{0\}$.

The overlap maps are:

$$\varphi_2 \circ \varphi_1^{-1}(z) = \frac{1}{z}, \qquad \varphi_1 \circ \varphi_2^{-1}(w) = \frac{1}{w}.$$

These are biholomorphic, since the inverse of $z \mapsto 1/z$ is again $z \mapsto 1/z$.

Thus, this defines a valid atlas, and the resulting Riemann surface is the **Riemann sphere**.

4.2.1 Stereographic Projection and the Metric

We identify the Riemann sphere with the unit sphere $S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$. Think of $\mathbb{C}$ as the plane $(x_1, y_1, 0)$. Given $z = x + iy \in \mathbb{C}$, draw the line from the north pole $(0, 0, 1)$ to $(x, y, 0)$.
This intersects the sphere at:

$$\begin{aligned} x_1 &= \frac{2x}{|z|^2 + 1} = \frac{2x}{x^2 + y^2 + 1}, \\ x_2 &= \frac{2y}{|z|^2 + 1} = \frac{2y}{x^2 + y^2 + 1}, \\ x_3 &= \frac{|z|^2 - 1}{|z|^2 + 1} = \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1}. \end{aligned}$$

Define

$$\sigma : S^2 \to \widehat{\mathbb{C}}$$

by

$$\sigma(x_1, x_2, x_3) = \begin{cases} \dfrac{x_1}{1 - x_3} + i\,\dfrac{x_2}{1 - x_3}, & (x_1, x_2, x_3) \neq (0, 0, 1), \\ \infty, & (x_1, x_2, x_3) = (0, 0, 1). \end{cases}$$

$$\sigma^{-1}(z) = \begin{cases} \left(\dfrac{2\Re z}{|z|^2 + 1}, \dfrac{2\Im z}{|z|^2 + 1}, \dfrac{|z|^2 - 1}{|z|^2 + 1} \right), & z \in \mathbb{C}, \\ (0, 0, 1), & z = \infty. \end{cases}$$

The spherical (or chordal) metric is:

$$d_{\widehat{\mathbb{C}}}(z, w)| = \frac{2|z - w|}{\sqrt{(1 + |z|^2)(1 + |w|^2)}},$$

with the natural extension when one point is ∞.

Chapter 5

Univalent Functions

5.1 Basic Principles

5.1.1 Recalling stuff that we already know?

Definition 5.1.

- A *domain* = open connected subset of $\mathbb{C}$.
- *Simply connected* is defined via the complement being connected (rather than "no holes" or "every loop contractible" these are equivalent for domains in $\mathbb{C}$, but the complement-in-$\hat{\mathbb{C}}$ characterization is the cleanest one to work with).
- A **neighborhood of a set** $E \subset \mathbb{C}$ is an open set which contains E.
- A **compact Set** is a *closed* and *bounded* subset of $\mathbb{C}$.
- The **closure** $\overline{E}$ of a set $E \subset \mathbb{C}$ is the smallest closed set that contains E. Equivalently, $\overline{E} = E \cup \{\text{limit points of } E\}$.
- The**Interior** E° of a set E is the largest open set contained in E. It may happen that $E^{\circ} = \emptyset$.
- The **boundary** of E is the set $\partial E = \overline{E} - E^{\circ}$, consisting of all points in the closure of E that are not interior points.

Notation: The unit disk $\mathbb{D} = \{|z| < 1\}$ and its boundary circle $\mathbb{T} = \{|z| = 1\}$.

Definition 5.2.

- An **Arc** in the complex plane is the continuous image of a line segment. Formally, an arc is a continuous mapping $z = \varphi(t)$ of an interval $a \leq t \leq b$ into $\mathbb{C}$.
- A **Rectifiable Arc** is an arc that has a finite length — that is, the mapping function φ is of *bounded variation*.

- A **Jordan Arc** is an arc without self-intersections.
- A **Closed Curve** is the continuous image of a circle, or equivalently, an arc whose endpoints coincide.
- A **Simple Closed Curve (Jordan Curve)** is a closed curve without "'

Theorem 5.1 (The Jordan Curve Theorem)**.** *The Jordan curve theorem asserts that every Jordan curve divides the plane into two regions the* interior *and the* exterior *of the curve.*

Core complex analysis, compressed:

- Analytic at $z_0 \Rightarrow$ infinitely differentiable there $\Rightarrow$ local Taylor series. This equivalence (holomorphic = analytic) is "one of the miracles" it has no real-variable analogue.
- Cauchy's integral formula $f(z) = \frac{1}{2\pi i}\oint_C \frac{f(\zeta)}{\zeta - z} d\zeta$ generates the Taylor coefficients by differentiating under the integral sign.
- **Uniqueness principle**: zeros of a nonzero analytic function are isolated (no accumulation point inside the domain).
- **Morera's theorem**: the converse of Cauchy vanishing of all contour integrals $\Rightarrow$ analytic. This is your go-to tool for proving analyticity of things defined by integrals (you'll see this trick again with Green's function below).
- **Maximum modulus principle**: $|f|$ has no interior local max unless f is constant "a peakless landscape," as Duren nicely puts it.
- **Schwarz lemma**: $f : \mathbb{D} \to \mathbb{D}$, $f(0) = 0 \Rightarrow |f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality only for rotations $f(z) = e^{i\theta} z$. Proof is one line: apply max modulus to $g(z) = f(z)/z$ (removable singularity at 0). This lemma is the seed of an enormous amount of univalent function theory half the book grows out of clever applications of it.
- **Residue theorem** and its corollary the **argument principle**: the number of zeros of f inside C equals $\frac{1}{2\pi i}\oint_C \frac{f'}{f} dz$, i.e. the winding number of $f(C)$ about 0. This is the "count zeros by watching the image curve wind around the origin" idea.
- **Rouché's theorem**: if $|g| < |f|$ on C, then f and $f + g$ have the same number of zeros inside C. Proof is slick: $\Delta_C \arg(f + g) = \Delta_C \arg f + \Delta_C \arg(1 + g/f)$, and the second term is 0 because $1 + g/f$ stays in the right half-plane (never winds around 0) when $|g/f| < 1$.
- **Hurwitz's theorem**: locally uniform limits of zero-free (or non-vanishing) analytic functions are either identically zero or zero-free themselves more precisely, zeros of the limit are limits of zeros of the f_n. Proved directly from Rouché by shrinking a circle around a zero z_0 of f and comparing f to f_n there.
- **Univalence is essentially closed under locally-uniform limits**: if $f_n \to f$ locally uniformly and each f_n is univalent, f is univalent *or constant* (the "or constant" is unavoidable e.g. $f_n(z) = z/n \to 0$). This

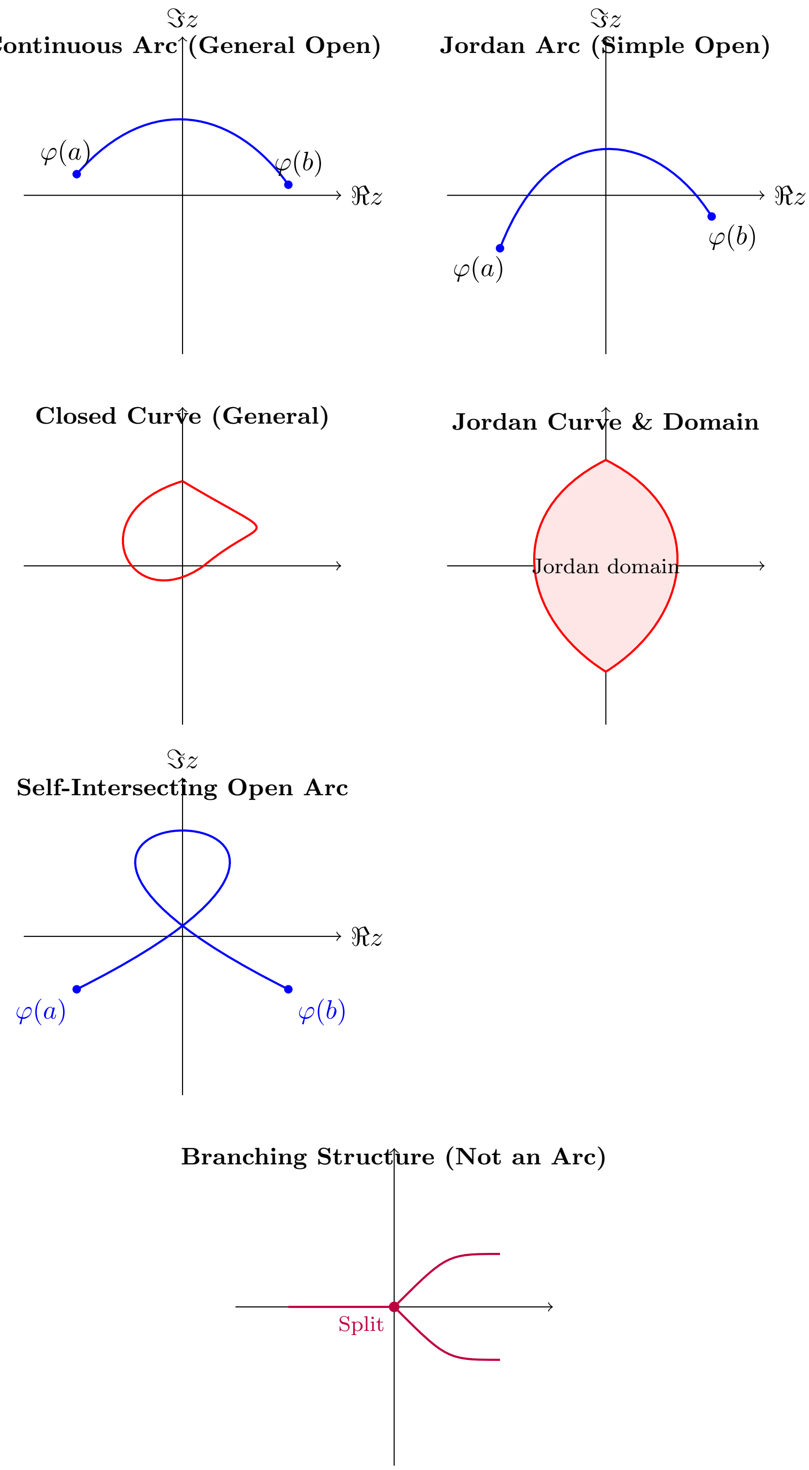


Figure 5.1: Comprehensive comparison of curves, arcs, and non-arc structures in the complex plane.

single theorem is why families of univalent functions behave so well under limits, and it's the linchpin for the compactness argument used later to prove Riemann mapping.

5.2 Local Mapping Properties

Since analytic $f = u + iv$ satisfies Cauchy–Riemann, the real Jacobian of the map $(x, y) \mapsto (u, v)$ works out to be exactly $|f'(z)|^2$. So:

- $f'(z_0) \neq 0 \iff f$ locally univalent near z_0 (inverse function theorem one way, Rouché the other way).
- **Warning example**: local univalence everywhere does *not* imply global univalence. $f(z) = z^2$ on the annular sector $1 < |z| < 2,\ 0 < \arg z < 3\pi/2$ is locally univalent throughout but not globally (angle range exceeds π, so some values get hit twice). This is exactly the kind of subtlety that matters for quadratic differentials and slit mappings local vs. global univalence is a recurring theme in your own extremal-problem work.

Geometric meaning of f':

- $|f'|^2$ = local area magnification $\Rightarrow$ Area of $f(D) = \iint_D |f'(z)|^2\, dx\, dy$.
- $|f'|$ = local arclength magnification $\Rightarrow$ length of $f(C) = \int_C |f'(z)|\, |dz|$.
- $\arg f'(z_0)$ = local rotation angle. This is why analytic univalent maps are called **conformal**: the rotation is the *same* for every arc through z_0, so the angle between two arcs is preserved.

Then: linear fractional (Möbius) transformations map "circles" (circles-or-lines) to "circles," preserve inverse-point pairs, and the automorphisms of $\mathbb{D}$ are exactly $f(z) = e^{i\theta} \frac{z-\alpha}{1-\bar{\alpha}z}$, $|\alpha| < 1$ derived from the Schwarz lemma.

5.3 Normal Families

- **Normal family**: every sequence has a locally-uniformly-convergent subsequence (this is the complex-analytic analogue of Bolzano–Weierstrass).
- **Local boundedness** (uniformly bounded on compacts) $\Rightarrow$ (via Cauchy's formula) the derivatives are also locally bounded.
- **Montel's theorem**: locally bounded $\Rightarrow$ normal. Proved by a diagonal argument: converge on a countable dense set, then use local boundedness (via the Cauchy estimate) to upgrade pointwise convergence on a dense set to uniform convergence on compacts essentially Arzelà–Ascoli done by hand.

- Converse also holds: normal ⇒ locally bounded.
- **Vitali's theorem**: for a normal family, convergence at just a set of points *with an accumulation point* in D already forces full locally-uniform convergence. (Proof: any two subsequential limits must agree on the accumulation set, hence agree everywhere by uniqueness.)

Punchline for the book: the class S (univalent, $f(0) = 0$, $f'(0) = 1$ on $\mathbb{D}$) is locally bounded via the growth theorem $|f(z)| \leq |z|(1 - |z|)^{-2}$ (proved later, in Ch. 2 in Duren[]), hence normal by Montel, hence (combining with the §1.1 univalence-limit theorem, ruling out the constant case since $f_n'(0) = 1$ forces $f'(0) = 1 \neq 0$) S **is a compact normal family**. This single fact is *the* reason extremal problems over S (like the ones you've been chasing with the $a_3 - \lambda a_2^2$ functional) are guaranteed to have solutions.

5.4 Extremal Problems

General machinery: a continuous functional ϕ on a compact normal family $\mathcal{F}$ always attains its supremum of $\operatorname{Re}\phi$ take a maximizing sequence, extract a locally-uniformly-convergent subsequence (compactness), use continuity of ϕ to pass to the limit. This is the abstract shape of *every* extremal-function existence proof in the book (and, not coincidentally, of the Schiffer-variation existence arguments you've been working through for Pfluger's paper).

Also: if $\mathcal{G}$ is dense in $\mathcal{F}$, sup over $\mathcal{G}$ = sup over $\mathcal{F}$ useful for restricting to "nice" functions (e.g. slit mappings) without changing the extremal value.

5.5 The Riemann Mapping Theorem

Koebe's compactness proof (not Riemann's original Dirichlet-principle argument, which had a gap Riemann didn't close):

1. Let $\mathcal{F}$ = univalent $f : D \to \mathbb{D}$ with $f(\zeta) = 0$, $f'(\zeta) > 0$. Show $\mathcal{F} \neq \emptyset$ (using simple connectivity to take a square-root branch of $z - \alpha$ for an omitted point α, then a Möbius map to squeeze into $\mathbb{D}$).
2. $\mathcal{F}$ is normal (Montel, since it's bounded by 1).
3. Maximize $f'(\zeta)$ over $\mathcal{F}$ extremal function f exists by the §1.4 machinery, and it's not constant since $f'(\zeta) = M > 0$.
4. Show f must be *onto* $\mathbb{D}$: if it missed a point ω, you can construct (via a square root of a Möbius transform of f) another candidate $G \in \mathcal{F}$ with $G'(\zeta) > f'(\zeta)$ contradicting extremality. This is the clever "square-root trick" step; it's the crux of the whole proof.
5. Uniqueness: any two such maps differ by an automorphism of $\mathbb{D}$ fixing 0 with positive derivative there, which is forced to be the identity.

Plus **Carathéodory's extension theorem**: if D is a Jordan domain, the Riemann map extends to a homeomorphism of the closures stated without proof (referred to Goluzin).

5.6 Analytic Continuation

- Basic continuation across a shared boundary arc where two analytic functions agree proved via a clever splitting of the Cauchy integral into two pieces that overlap correctly on either side.
- Upgraded version: continuity across the arc suffices, analyticity *on* the arc isn't needed a priori (this is the more useful, harder version, proved via the same integral-splitting trick using an "extended Cauchy theorem" for continuous-on-boundary functions).
- **Schwarz reflection principle** (special case): if f is analytic in $\mathbb{D}$, continuous up to a boundary arc Γ, and real on Γ, then $f(z) := \overline{f(1/\bar{z})}$ for $|z| > 1$ analytically continues f across Γ.
- **General reflection** across analytic arcs mapping to analytic arcs: reduce to the real-line case via local analytic parametrizations of both arcs, conjugate by these parametrizations, apply the basic reflection principle, then transport back. Key upshot for univalent function theory: **a function in S mapping onto a Jordan domain with analytic boundary extends univalently across $|z| = 1$ to a slightly larger disk** this fact gets used constantly later (e.g. to justify boundary behavior of extremal functions in coefficient problems).

5.7 Harmonic and Subharmonic Functions

- States **Green's theorem/formula** $\oint_C (\phi\,\partial\psi/\partial n - \psi\,\partial\phi/\partial n)\,ds = \iint_D (\phi\,\Delta\psi - \psi\,\Delta\phi)\,dx\,dy$ this is the workhorse used again in §1.8 to prove Green's function is symmetric.
- Real/imaginary parts of analytic functions are harmonic (Cauchy–Riemann $\Rightarrow$ Laplace's equation); harmonic conjugates exist locally (globally on simply connected domains, via a path integral that's path-independent by Green's theorem).
- Harmonic $\circ$ analytic = harmonic.
- **Mean value property**: $u(z_0) = \frac{1}{2\pi}\int_0^{2\pi} u(z_0 + re^{i\theta})\,d\theta$, derived from Green's theorem applied to ∇u.
- **Poisson formula**: reconstructs u inside a disk from boundary values, via the Poisson kernel $P_r(\theta) = \frac{R^2-r^2}{R^2-2Rr\cos\theta+r^2}$.
- **Maximum principle for harmonic functions**, and the **converse mean-value characterization**: continuous + local mean-value property $\Rightarrow$ harmonic (proved by comparing u to its Poisson integral U and using the maximum principle on $u - U$).

- **Subharmonic functions**: $u \le U$ (Poisson integral) on every disk; equivalently, continuous + local *sub*-mean-value property. Key facts: sums of subharmonic functions are subharmonic; $\max(u, v)$ is subharmonic; convex increasing functions of subharmonic functions are subharmonic (via Jensen's inequality) this is exactly how you get $|u|^p$ ($p \ge 1$, u harmonic) and $|f|^p$ ($p > 0$, f analytic) subharmonic, and $\log |f|$ subharmonic in the generalized sense (allowing $-\infty$) facts used throughout the coefficient-estimate machinery in later chapters.

5.8 Green's Functions

For a finitely-connected domain D with smooth boundary, Green's function is

$$g(z, \zeta) = h(z, \zeta) - \log |z - \zeta|,$$

where $h(\cdot, \zeta)$ is the harmonic function solving the Dirichlet problem with boundary data $\log |z-\zeta|$ (so $g = 0$ on the boundary and g has the right log singularity at ζ).

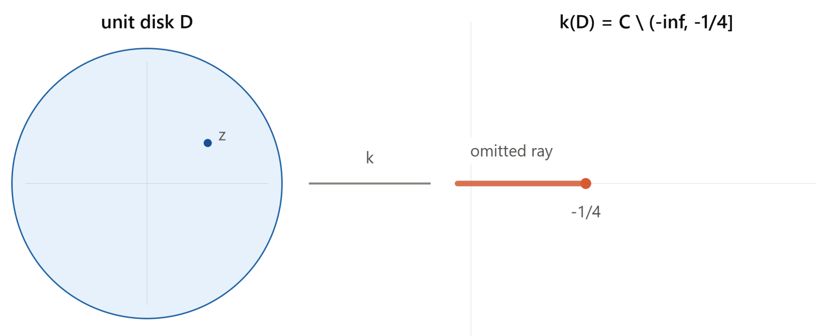
- **Positivity**: $g > 0$ inside, by the maximum principle.
- **Symmetry** $g(z, \zeta) = g(\zeta, z)$: proved by excising small ϵ-disks around two points ζ_1, ζ_2, applying Green's formula to $g(\cdot, \zeta_1)$ and $g(\cdot, \zeta_2)$ on the resulting domain, and letting $\epsilon \to 0$ the log singularities' normal derivatives blow up in exactly the right way ($\sim 1/\epsilon$) to produce the residue-like terms $-2\pi g(\zeta_1, \zeta_2) + 2\pi g(\zeta_2, \zeta_1) = 0$.
- **Reproducing formula**: for harmonic u,

$$u(\zeta) = -\frac{1}{2\pi} \int_C u \, \frac{\partial g}{\partial n} \, ds,$$

  generalizing the Poisson formula to arbitrary domains same excise-and-let-$\epsilon \to 0$ technique.
- **Connection to the Riemann map**: if $f : D \to \mathbb{D}$ is the Riemann map with $f(\zeta) = 0$, then $g(z, \zeta) = -\log |f(z)|$. Conversely, knowing g recovers f via $f(z) = \exp\{-g(z, \zeta) - i\, \tilde{g}(z, \zeta)\}$ where $\tilde{g}$ is the harmonic conjugate (well-defined mod 2π around ζ, so f is still single-valued). This Green's-function $\leftrightarrow$ conformal-map duality is precisely the language your Nehari/Schippers reading has been using (period matrices, canonical domains) it's the classical one-variable shadow of the multiply-connected constructions you've been chewing on.

Chapter 6

Elementary Theory of Univalent Functions



Introduction — the class S

Univalence. f is *univalent* (schlicht) on a domain D if $f(z_1) = f(z_2) \implies z_1 = z_2$. For analytic f, *local* univalence at z_0 is equivalent to $f'(z_0) \neq 0$. An analytic univalent map is automatically a conformal mapping (angle-preserving), since $f'(z_0) \neq 0$ means the map looks like a rotation-and-scaling infinitesimally.

The normalized class. S consists of f analytic and univalent on $\mathbb{D} = \{|z| < 1\}$ with

$$f(0) = 0, \qquad f'(0) = 1,$$

so every $f \in S$ has a Taylor expansion

$$f(z) = z + a_2 z^2 + a_3 z^3 + \cdots, \qquad |z| < 1.$$

By the Riemann mapping theorem, this normalized disk-based class is essentially universal: any univalent map of a simply connected domain (with ≥ 2 boundary points) onto another can be built from a member of S by pre- and post-composing with Möbius maps.

The Koebe function is the star example:

$$k(z) = \frac{z}{(1-z)^2} = z + 2z^2 + 3z^3 + \cdots = \sum_{n=1}^{\infty} n z^n.$$

To see its image, write

$$k(z) = \frac{1}{4}\left(\left(\frac{1+z}{1-z}\right)^2 - 1\right).$$

The Möbius map $\zeta = \dfrac{1+z}{1-z}$ sends $\mathbb{D}$ conformally onto the right half-plane $\operatorname{Re}\zeta > 0$. Squaring a right half-plane sweeps it around into the *entire plane minus the nonpositive reals* $(-\infty, 0]$, and then $\frac{1}{4}(\zeta^2 - 1)$ shifts and scales that slit to $(-\infty, -\frac{1}{4}]$. That's exactly the picture in the diagram: k maps $\mathbb{D}$ **onto** $\mathbb{C}$ **minus the ray** $(-\infty, -1/4]$**.**

Other examples in S:

f	image
z	$\mathbb{D}$ itself
$z(1-z)^{-1}$	half-plane $\operatorname{Re} w > -\frac{1}{2}$
$z(1-z^2)^{-1}$	plane minus two rays $[\frac{1}{4}, \infty)$ and $(-\infty, -\frac{1}{4}]$
$\frac{1}{4}\log\frac{1+z}{1-z}$	horizontal strip $\lvert\operatorname{Im} w\rvert < \pi/4$
$z - \frac{1}{2}z^2 = \frac{1}{4}(1-(1-z)^2)$	interior of a cardioid

S is not a vector space. The sum of two functions in S need not be univalent — e.g. $\dfrac{z}{1-z} + \dfrac{z}{1+iz}$ has a critical point inside $\mathbb{D}$. This is a good reminder that S is a *geometric* class (defined by injectivity), not an algebraic one.

Symmetries preserving S. These are the standard moves you'll use constantly:

1. **Conjugation:** $g(z) = \overline{f(\bar{z})} \in S$.
2. **Rotation:** $g(z) = e^{-i\theta} f(e^{i\theta} z) \in S$.
3. **Dilation:** $g(z) = r^{-1} f(rz) \in S$ for $0 < r < 1$.
4. **Disk automorphism:** for $|\zeta| < 1$, $g(z) = \dfrac{f\left(\frac{z+\zeta}{1+\bar{\zeta}z}\right) - f(\zeta)}{(1-|\zeta|^2)f'(\zeta)} \in S$. This is the workhorse move used in the proofs of the distortion and growth theorems below — it re-centers f at an arbitrary point ζ while renormalizing back to class S.

5. **Range transformation:** if ψ is univalent on $f(\mathbb{D})$ with $\psi(0) = 0, \psi'(0) = 1$, then $\psi \circ f \in S$.
6. **Omitted-value transformation:** if f omits the value w, then $g = \dfrac{wf}{w-f} \in S$ (this is used directly in the Koebe 1/4 theorem below).
7. **Square-root transformation:** if $f \in S$, then

$$g(z) = \sqrt{f(z^2)} = z + \tfrac{1}{2}a_2 z^3 + \cdots \in S.$$

Why (7) is univalent — a nice small argument. Since $f(w) = 0$ only at $w = 0$, we can pick a single-valued branch $g(z) = \sqrt{f(z^2)}$; g is automatically *odd*, $g(-z) = -g(z)$. If $g(z_1) = g(z_2)$, squaring gives $f(z_1^2) = f(z_2^2)$, so $z_1^2 = z_2^2$ (univalence of f), i.e. $z_1 = \pm z_2$. If $z_1 = -z_2$, then $g(z_1) = g(z_2) = g(-z_1) = -g(z_1)$, forcing $g(z_1) = 0$, hence $z_1 = 0 = z_2$. Either way $z_1 = z_2$. ■

m-fold symmetric functions. $S^{(m)} \subset S$ consists of functions with m-fold symmetry, $f(z) = z + \sum_{k\geq 1} a_{mk+1} z^{mk+1}$. The m-th root transform $g(z) = \{f(z^m)\}^{1/m}$ sends $S \to S^{(m)}$, and this correspondence is onto: every $g \in S^{(m)}$ arises this way. (The square-root transformation above is the case $m = 2$.)

The exterior class Σ. This is the natural companion class: functions

$$g(z) = z + b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \cdots$$

univalent on $\Delta = \{|z| > 1\}$ except for a simple pole at ∞ (residue 1). Each $g \in \Sigma$ maps Δ onto the complement of some compact connected set E.

- $\Sigma' \subset \Sigma$: those g with $0 \notin g(\Delta)$, i.e. g never vanishes. Any $g \in \Sigma$ can be shifted into Σ' by adjusting b_0 (a translation, which preserves univalence).
- **Inversion**, $f \mapsto g$: for $f \in S$,

$$g(z) = \{f(1/z)\}^{-1} = z - a_2 + (a_2^2 - a_3)\frac{1}{z} + \cdots \in \Sigma',$$

  and this is a bijection $S \leftrightarrow \Sigma'$.
- Σ' is closed under a square-root transform $G(z) = \sqrt{g(z^2)}$ — but (important caveat) this only works starting from Σ', not general Σ: if g has a zero, $g(z^2) = 0$ somewhere creates a branch point and the square root can't be made single-valued.
- $\Sigma_0 \subset \Sigma$: the subclass with $b_0 = 0$ (achievable by translation, though generally *not* simultaneously with membership in Σ').
- $\tilde{\Sigma}$: "full mappings," those $g \in \Sigma$ whose omitted set E has 2-dimensional Lebesgue measure zero.

6.1 The Area Theorem

This is the load-bearing result of the whole chapter — nearly everything downstream is a corollary.

> **Theorem 2.1 (Area Theorem, Gronwall 1914).** If $g \in \Sigma$, $g(z) = z + b_0 + \sum_{n \geq 1} b_n z^{-n}$, then
>
> $$\sum_{n=1}^{\infty} n|b_n|^2 \leq 1,$$
>
> with equality iff $g \in \tilde{\Sigma}$.

Proof idea (this is why it's called the "area" theorem). Fix $r > 1$ and let C_r be the image of $|z| = r$ under g; since g is univalent, C_r is a simple closed curve enclosing a region $E_r \supset E$ (the omitted set). By Green's theorem, the enclosed area is

$$\text{area}(E_r) = \frac{1}{2i} \oint_{C_r} \bar{w}\, dw.$$

Parametrize $w = g(re^{i\theta})$ and expand: because $g(z) = z + b_0 + \sum b_n z^{-n}$, carrying out $\bar{w}\, dw$ and integrating θ from 0 to 2π kills all cross terms by orthogonality of $e^{ik\theta}$, leaving

$$\text{area}(E_r) = \pi\Big(r^2 - \sum_{n=1}^{\infty} n|b_n|^2 r^{-2n}\Big), \qquad r > 1.$$

Since area is ≥ 0 for every $r > 1$, let $r \downarrow 1$: the outer measure of E satisfies

$$m(E) = \pi\Big(1 - \sum n|b_n|^2\Big) \geq 0,$$

which is exactly the theorem. Equality $m(E) = 0$ is precisely membership in $\tilde{\Sigma}$. ■

This is a genuinely lovely argument: you get a *coefficient inequality* out of nothing but "area can't be negative."

Corollary. $|b_1| \leq 1$, with equality iff $g(z) = z + b_0 + b_1/z$, $|b_1| = 1$ — mapping Δ onto the complement of a line segment of length 4. (Equality in $|b_1|^2 \leq \sum n|b_n|^2 \leq 1$ forces all other $b_n = 0$ and $|b_1| = 1$.)

Bieberbach's Theorem (1916), via the corollary.

> **Theorem 2.2.** If $f \in S$, then $|a_2| \leq 2$, with equality iff f is a rotation of the Koebe function.

Proof. Chain the square-root transform and inversion: $g(z) = \{f(1/z^2)\}^{-1/2} = z - \frac{1}{2}a_2 z^{-1} + \cdots \in \Sigma$. Its b_1 coefficient is $-a_2/2$, so the corollary gives $|a_2/2| \leq 1$, i.e. $|a_2| \leq 2$. Equality forces $g(z) = z + b_0 + \frac{b_1}{z}$ with $|b_1| = 1$, and unwinding the transform shows this corresponds to f being a rotation $e^{-i\theta}k(e^{i\theta}z)$ of the Koebe function. ■

Koebe's One-Quarter Theorem — this is arguably the single most quoted fact in the subject.

Theorem 2.3. Every $f \in S$ has range containing the disk $|w| < \frac{1}{4}$.

Proof. Suppose f omits a value w. Apply the omitted-value transformation:

$$g(z) = \frac{wf(z)}{w - f(z)} = z + \left(a_2 + \frac{1}{w}\right) z^2 + \cdots \in S.$$

Bieberbach's theorem applied to g gives $\left|a_2 + \dfrac{1}{w}\right| \leq 2$. Combined with $|a_2| \leq 2$ (triangle inequality: $\left|\frac{1}{w}\right| \leq \left|a_2 + \frac{1}{w}\right| + |a_2| \leq 4$), we get $|w| \geq \frac{1}{4}$. ■

The proof shows more: **only** the Koebe function and its rotations actually achieve $|w| = \frac{1}{4}$ (they're the ones omitting a boundary point of the disk $|w| < 1/4$ exactly) — every other $f \in S$ covers a strictly larger disk. Also worth noting: univalence is essential here, not just $f(0) = 0, f'(0) = 1$ — $f_n(z) = z(1 + z/n)^2$ satisfies the normalization but omits values arbitrarily close to 0, since it's not univalent.

6.2 Growth and Distortion Theorems

These translate the coefficient bound $|a_2| \leq 2$ into pointwise geometric control on f itself.

Theorem 2.4 (the key technical estimate). For $f \in S$, $|z| = r < 1$,

$$\left|\frac{zf''(z)}{f'(z)} - \frac{2r^2}{1 - r^2}\right| \leq \frac{4r}{1 - r^2}.$$

Proof sketch. Fix $\zeta \in \mathbb{D}$ and apply the disk-automorphism symmetry (item 4 above) to build $F \in S$ re-centered at ζ. Expanding F's second Taylor coefficient in terms of f's Taylor coefficients at ζ gives an expression $A_2(\zeta) = \frac{1}{2}(1 - |\zeta|^2)\dfrac{f''(\zeta)}{f'(\zeta)} - \bar{\zeta}$. Bieberbach's theorem bounds $|A_2(\zeta)| \leq 2$; simplifying and relabeling $\zeta \to z$ gives the stated inequality. ■

Splitting into real/imaginary parts of this one inequality produces *three* classical theorems by integrating along radii — this is a nice piece of technique worth internalizing (it's a recurring move in the subject: one sharp pointwise bound on a logarithmic-derivative-type quantity, then radial integration, gives sharp global bounds).

Distortion Theorem 2.5. Take real parts: $\operatorname{Re}\left(\dfrac{zf''}{f'}\right)$ is sandwiched, and since $\operatorname{Re}\dfrac{zf''}{f'} = r\dfrac{\partial}{\partial r}\log|f'(re^{i\theta})|$ (using a branch of $\log f'$ vanishing at 0), integrating in r gives

$$\frac{1 - r}{(1 + r)^3} \leq |f'(z)| \leq \frac{1 + r}{(1 - r)^3}, \qquad |z| = r,$$

with equality (for $z \neq 0$) iff f is a rotation of k. Both bounds are attained by $k'(z) = \dfrac{1+z}{(1-z)^3}$ itself.

Growth Theorem 2.6. Integrating $f(z) = \int_0^z f'$ radially and applying the distortion theorem along the way:

$$\frac{r}{(1+r)^2} \leq |f(z)| \leq \frac{r}{(1-r)^2}, \qquad |z| = r,$$

again with equality only for rotations of the Koebe function. The *lower* bound is the subtler direction: if $|f(z)| < \frac{1}{4}$, the Koebe 1/4 theorem guarantees the whole segment from 0 to $f(z)$ lies in $f(\mathbb{D})$, so you can integrate f' along its (univalent) preimage arc and apply the distortion theorem there.

Rotation theorem (imaginary part). Taking imaginary parts of Theorem 2.4 instead and integrating gives the (non-sharp) bound

$$|\arg f'(z)| \leq 2 \log \frac{1+r}{1-r}.$$

The genuinely **sharp** rotation theorem,

$$|\arg f'(z)| \leq \begin{cases} 4 \arcsin r, & r \leq 1/\sqrt{2}, \\ \pi + \log \dfrac{r^2}{1-r^2}, & r > 1/\sqrt{2}, \end{cases}$$

lies much deeper (needs Loewner's method, Chapter 3) — and the fact that the sharp bound has *different functional forms* on either side of $r = 1/\sqrt{2}$ is flagged as one of the striking phenomena of the theory.

Theorem 2.7 (combined growth/distortion, for zf'/f). For $f \in S$,

$$\frac{1-r}{1+r} \leq \left| \frac{zf'(z)}{f(z)} \right| \leq \frac{1+r}{1-r}.$$

Note this does **not** follow by naively dividing the growth and distortion bounds — the proof has to go back to the auxiliary function F from Theorem 2.4's construction and use the growth theorem on F at the point $-\zeta$. Equality, again, only for rotations of k, for which $\dfrac{zk'(z)}{k(z)} = \dfrac{1+z}{1-z}$ exactly.

6.3 Coefficient Estimates

The Bieberbach Conjecture. Since k has n-th coefficient exactly n, and $|a_2| \leq 2$ with the same extremal function, it's natural to conjecture $|a_n| \leq n$ for all $f \in S$, with equality only for rotations of k. At the time this book was written, this was proven only for small n — it's worth flagging for you: **de**

Branges proved the full conjecture in 1985, after this book's publication (using Loewner theory plus special-function/hypergeometric estimates related to Askey–Gasper). Duren's book predates that, so what follows is the state of the art *before* de Branges.

Finiteness of $A_n = \sup_{f \in S} |a_n|$ follows either from compactness of S in the topology of locally uniform convergence (Montel-type argument, plus continuity of a_n as a functional), or crudely from Cauchy's estimate combined with the growth theorem, giving $A_n \le Cn^2$.

Littlewood's Theorem (1925). $|a_n| \le en$ for all $f \in S$, $n \ge 2$ — the correct *order of magnitude*, proved by completely elementary means.

The proof rests on an integral-means lemma. Define $M_p(r, f) = \left(\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^p \, d\theta \right)^{1/p}$.

> **Lemma.** For $f \in S$, $M_1(r, f) \le \dfrac{r}{1-r}$, $0 < r < 1$.

Proof. Let $h(z) = \sqrt{f(z^2)} = z + e_3 z^3 + e_5 z^5 + \cdots$ (the square-root transform, an odd function in S). By the growth theorem applied to f at the point z^2 (modulus r^2), $|f(z^2)| \le r^2/(1-r^2)^2$, so

$$|h(z)| \le \frac{r}{1-r^2}, \qquad |z| = r.$$

So h maps $|z| < r$ into the disk $|w| < r/(1-r^2)$; the (multiplicity-weighted) area of the image is

$$A_r = \iint_{|z|<r} |h'(z)|^2 \, dA = \pi \sum_n n |e_n|^2 r^{2n} \;\le\; \pi \frac{r^2}{(1-r^2)^2}.$$

Dividing by r and integrating from 0 to r:

$$\sum_n |e_n|^2 r^{2n} \le \frac{r^2}{1-r^2}.$$

By Parseval, the left side is $M_2(r,h)^2$, and since $|h(z)|^2 = |f(z^2)|$, a change of variable shows $M_2(r,h)^2 = M_1(r^2, f)$. Relabeling $r^2 \to r$ gives $M_1(r,f) \le r/(1-r)$. ■

Deriving Littlewood's theorem from the lemma. Cauchy's formula gives $|a_n| \le r^{-n} M_1(r,f) \le \dfrac{r^{1-n}}{1-r}$. Minimizing over r at $r = 1 - 1/n$:

$$|a_n| \le n \left(1 - \frac{1}{n}\right)^{1-n} \longrightarrow en \quad (\text{since } (1-1/n)^{-n} \to e).$$

How sharp is this? Not very — even in principle, this *method* (Cauchy estimate + integral means) can't beat $\frac{e}{2}n$, because the true extremal is the Koebe function, whose own integral mean can be computed in closed form:

$$M_1(r,k) = \frac{1}{2\pi}\int_0^{2\pi} \frac{r\,d\theta}{|1-re^{i\theta}|^2} = \frac{r}{1-r^2}$$

(a Poisson-kernel identity), which is smaller than Littlewood's elementary bound $r/(1-r)$ by essentially a factor of 2 near $r=1$ (since $1-r^2 \approx 2(1-r)$). It took until **Baernstein (1973)** — using his theory of "star-functions" — to prove the *sharp* statement $M_p(r,f) \le M_p(r,k)$ for **all** $f \in S$ and all $0 < p < \infty$, matching the Koebe function exactly. That recovers the optimal constant $\frac{e}{2}$ from this style of argument (though it still falls short of de Branges's exact n).

Application: arclength of image circles. Let $L_r(f)$ be the length of the image of $|z| = r$ under f, i.e. $L_r(f) = \int_0^{2\pi} |f'(re^{i\theta})|\,r\,d\theta$.

Theorem 2.9. $L_r(f) \le \dfrac{2\pi r(1+r)}{(1-r)^2}$.

Proof. Write $|f'(z)|\,r = \left|\dfrac{zf'(z)}{f(z)}\right| \cdot |f(z)|$ (using $|z| = r$), bound the first factor by Theorem 2.7's upper bound $\frac{1+r}{1-r}$, and integrate:

$$L_r(f) \le \frac{1+r}{1-r}\int_0^{2\pi} |f(re^{i\theta})|\,d\theta = \frac{1+r}{1-r}\cdot 2\pi M_1(r,f) \le \frac{1+r}{1-r}\cdot 2\pi \cdot \frac{r}{1-r} = \frac{2\pi r(1+r)}{(1-r)^2}. \blacksquare$$

This is again "very nearly sharp" but not sharp; using Baernstein's inequality $M_1(r,f) \le \frac{r}{1-r^2}$ in place of the elementary lemma improves it to

$$L_r(f) \le \frac{2\pi r}{(1-r)^2}.$$

(The $(1+r)$ factor cancels exactly against the extra factor gained from the sharper M_1 bound.) The genuinely sharp bound for $L_r(f)$ over all of S is still **unknown** in general; Clunie and Duren later showed $L_r(f) \le L_r(k)$ holds within the subclass of *close-to-convex* functions (§2.6).

6.4 Where the excerpt trails off

The chapter then pivots (§2.5) into geometrically-defined subclasses — **convex functions** C (image is a convex domain) and **starlike functions** S^* (image starlike w.r.t. 0), with $C \subset S^* \subset S$, and the Koebe function itself is starlike but not convex. This sets up the class P of functions with positive real part (Carathéodory functions) via the Herglotz/Poisson–Stieltjes representation — the tool that will let you prove the Bieberbach conjecture *sharply* within these two subclasses, which is presumably where the excerpt continues next.

Chapter 7

06-Important Theorms

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Every locally bounded family of analytic functions is normal.
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Proof.

$\mathcal{F}$ = family of functions analytic in a domain D

$\phi : \in \mathcal{F}$, ϕ = complex valued functional

Ex: $\phi(f) = f(\xi)$ for fixed $\xi \in D$.

point evaluation map

Def$^{\underline{n}}$: Contiunity of a functional.

A futional ϕ is said to be <u>cts</u> if $\phi(f_n) \to \phi(f)$ whenever a sequnce of functions $f_n \in \mathcal{F}$ cgt uniformly on each compact subset D to a function $f \in \mathcal{F}$.

ex:- Point Evaluation ~~Map~~ functional. is cts

$\phi(f) = f(\xi)$. If $f_n \to f$ uniformly on compact subsets then one-point $\{\xi\}$ is compact ~~Here~~ $|f_n(\xi) - f(\xi)| \to 0$

Thus, $\phi(f_n) = f_n(\xi) \downarrow f(\xi) = \phi(f)$

- derviate of functional is $\phi(f) = f'(\xi)$ (By cauchy intergal formula)

~~explain~~ $$f_n'(\xi) = \frac{1}{2\pi i} \int_\gamma \frac{f_n(w)}{(w-\xi)^2}\, dw, \quad \gamma \text{ - small circle around } \xi$$

Since $f_n \to f$ uniformly on γ

$$\int_\gamma \frac{f_n(w)}{(w-\xi)^2}\, dw \longrightarrow \int_\gamma \frac{f(w)}{(w-\xi)^2}\, dw \Rightarrow \underset{\phi(f_n)}{\overset{}{f_n'(\xi)}} \to \underset{\phi(f)}{f'(\xi)}$$

“ “

□